

Petrol Pump Management

Complete Feature Documentation

Frappe 15 / ERPNext 15 Application

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56 DocTypes | 33 Reports | 7 Roles | 15 Scheduler Tasks

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1. Module Overview

Layer 1 - Configuration	Station setup, tank & nozzle config	Station Config, Tank Master, Nozzle Master, Fuel Price
Layer 2 - Operations	Shift, sales, stock management	Shift, Fuel Sale, Meter Reading, Daily Stock Register
Layer 3 - Credit & ANPR	Vehicle credit billing, ANPR	Vehicle Master, PP Customer, Credit Sale Invoice
Layer 4 - Compliance	GST, payroll, financial reports	GSTR-1/3B, Salary Slip, Expense Entry

2. Operational Workflows

2.1 Vehicle-based Credit Billing (ANPR)

Vehicle Arrival !' ANPR Camera captures plate !' OCR reads plate
!' Vehicle Master match (confidence check)
!' Credit Limit Ledger validation (available balance check)
!' Manager override if over limit
!' Fuel Sale created !' Auto Credit Sale Invoice generated
!' SMS/WhatsApp receipt sent !' Credit Points awarded
!' Credit Limit Ledger updated in real time

2.2 Shift & Fuel Sales Operations

Shift Start !' Manager allots nozzles + opening cash
!' Opening meter readings recorded per nozzle
!' Sales during shift (Cash / Credit / Card / UPI / Petro-card)
!' Closing meter readings taken
!' Tank dip check !' Compared against calculated stock
!' Shift settlement: Cash collected vs expected
!' Closing cash handed over !' Settlement approved

2.3 Stock & Tank Management

Purchase order !' Tanker arrival !' Trip Voucher created
!' Pre-unload dip & density recorded
!' Fuel decanted into tank !' Stock updated !' Variance flagged
!' Daily stock register auto-computed
!' Low-stock alerts triggered

2.4 Credit Recovery & Compliance

Statements auto-generated !' Overdue SMS/email reminders
!' Interest charged on late payments
!' Payments recorded against invoices
!' Bank reconciliation via CSV import
!' GST reports: GSTR-1, GSTR-3B, TCS, TDS

2.5 Employee & Payroll

Attendance marked !' Leave processed !' Overtime logged
!' Monthly salary slips auto-generated
!' Advances deducted !' Commission calculated

2.6 Digital Payments & Settlement

Card/UPI/E-Wallet transactions at POS
!' Swipe Settlement per machine !' Difference flagged

3. Implemented Features (56 DocTypes)

Configuration (8 DocTypes)

- Station Configuration - station name, licence, GST, currency
- Tank Master - fuel type, capacity, dip chart, stock level
- Tank Dip Chart - dip cm to volume mapping
- Nozzle Master - linked to tank, pump, status (Open/Stop/Reset)
- Fuel Price Master - effective date, rate, revision history
- Employee Master - role, salary type, mobile
- Commission Rule - per fuel type/qty thresholds
- PP Supplier Master

Operations (13 DocTypes)

- Shift - salesman assignment, opening cash, settlement
- Shift Nozzle Allotment - child table for nozzle assignments
- Fuel Sale - nozzle, quantity, rate, payment mode
- Meter Reading - opening/closing, testing qty, sale qty
- Daily Stock Register - opening, purchase, sale, closing
- Stock Purchase Decantation - invoiced vs received, density
- Trip Voucher - tanker trip tracking
- Lube Stock - lube inventory with expiry alerts
- Evaporation Loss - fuel loss tracking
- Tanker Expense - tanker trip P&L
- Bank Statement Import - CSV upload + auto-matching
- Station Inspection - dip, density, temperature
- Day Settlement - end-of-day reconciliation

Credit & Sales (9 DocTypes)

- PP Customer - type, GST, credit limit, points, risk
- Vehicle Master - plate, type, customer, ANPR status
- Credit Sale Invoice - amount, due date, interest, status
- Credit Statement - period-wise billing
- Credit Recovery Entry - discount, grace days, balance
- Credit Limit Ledger - limit, used, available, auto-block
- Payment Receipt - customer, mode, amount
- ANPR Scan Log - plate, camera, confidence
- Payment Receipt Invoice

Finance & HR (11 DocTypes)

- Expense Entry - date, type, amount, payment method
- Income Entry - type, source, amount, recurring
- Attendance Register - shift-wise, salary type
- Leave Application - employee, dates, status
- Overtime Log - hours, pay calculation
- Advance Amount - employee, amount, recovery
- Salary Slip Entry - basic, OT, commission, deductions
- Commission Payment - auto-calculated from rules
- Bank Deposit - amount, date, bank reference
- Cheque Print Queue - payee, amount, words
- Reward Points Ledger - earn/redeem/expire

Digital Payments (2 DocTypes)

- Swipe Settlement - machine, payment mode, collected
- Petro Card Transaction - card, amount, type

Compliance & GST (8 DocTypes)

- GSTR-1 Filing - B2B invoices, XML generation
- GSTR-3B Filing - auto-calculation, XML generation
- GSTR-2A Reconciliation - purchase matching
- TCS Statement - auto-calculation
- TDS Statement - auto-calculation
- Bank Reconciliation Entry
- Tally Export Log
- GSTR-1 Filing Invoice (child table)

Notifications (3 DocTypes)

- SMS Log - Twilio/MSG91/custom gateway
- Email Log - Frappe email system
- PP Notification Settings - gateway config + templates

4. Implemented Reports (33 Reports)

Daily Sales Summary	Shift-wise, nozzle-wise, payment-mode-wise sales
Shift Settlement Report	Cash collected vs expected, shortages
Stock Variation Report	Tank-wise dip vs meter-sale variance
Credit Customer Ageing	Outstanding dues by age bucket
ANPR Scan Report	Scan stats, auto-matched vs manual
GST VAT Summary	GST summary with CGST/SGST
Employee Attendance Payroll	Attendance, salary, commission
Fuel Rate Variation Report	Historical rate changes
Density Dip Variation	Tank-wise density vs stock
Profit Loss Statement	Trading account, expenses, margin
Vehicle Wise Consumption	Fuel consumption per vehicle
Swipe Digital Settlement	Card/UPI/e-wallet reconciliation
Expense Summary	Expense breakdown
Bank Deposit Report	Deposit history
Bank Reconciliation Report	Reconciliation status
GSTR-1 Summary	B2B credit sales for GST
GSTR-3B Summary	Monthly GST return
GSTR-2A Reconciliation Report	Purchase matching
HSN Wise Summary	HSN code-wise summary
TCS TDS Report	Combined TCS/TDS statement
Employee Commission Report	Employee-wise commission
Cash Flow Report	Daily cash inflows/outflows
PP Day Book	All transactions for a date
Payroll Summary	Employee-wise salary
Tally Export Report	Export history
Report Export	CSV/Excel export
Nozzle Wise Sale	Per-nozzle sales with split
Meter Dip Variation	Meter vs dip reading
Employee Shortage Report	Per-employee shortage
Monthly Shortage Report	Monthly aggregated shortage
Machine Wise Transaction	POS machine breakdown
Tank Valuation Report	Stock value & utilization
Sales Officer Inspection	Inspection records

5. Missing Features & Why They Cannot Be Implemented

Mobile Apps

Dealer / Owner mobile app	Requires React Native/Flutter - not Frappe
Manager-level mobile app	Needs native mobile app codebase
Salesman / DSM mobile app	Needs native mobile app codebase
Credit-customer self-service app	Full mobile app needs separate development

External Integrations

ANPR Camera / OCR	Requires physical camera + OCR API
Fuel Dispenser / Pump Controller	Requires hardware integration
Fleet / GPS Tracking	Requires GPS hardware + API
WhatsApp integration	Requires Business API + Meta approval
Payment Gateway	Requires merchant account + API keys
Bank auto-import	Banks dont offer open APIs
GST portal auto-submit	Requires DSC + GSTN registration
Tally real-time sync	Requires Tally Gateway (ODBC)

Minor Missing Features

Graphical dashboard	Needs manual UI configuration in Frappe Desk
Cheque printing layout	Requires bank-specific cheque format
Full accounting vouchers	Needs ERPNext accounting integration
Birthday DOB field	Field not yet added to PP Customer
Cash denomination breakdown	Not implemented as child table
Multi-outlet support	Single station only - needs architecture changes

6. Roles & Permissions

System Manager	Full access to all DocTypes
Station Manager	Full operations, read compliance
Salesman DSM	Own shift records, create fuel sales
Credit Accounts Officer	Full credit management
Compliance Officer	Full GST/compliance
Recovery Officer	Credit recovery management
Petrol Pump Auditor	Read-only across all DocTypes

7. Scheduler Tasks

Daily	Stock Reconciliation	Auto-compute daily stock register
Daily	Credit Reminders	SMS for overdue invoices
Daily	Stock Level Check	Alert for low stock
Daily	Auto-block Customers	Block on limit breach + SMS
Daily	Expiry Alerts	Lube expiry notifications
Daily	Expire Reward Points	Points older than 1 year
Daily	Evaporation Calc	Tank-wise evaporation loss
Daily	Business Summary SMS	SMS to manager
Daily	Birthday SMS	Birthday greetings
Weekly	Credit Statements	Auto-generate statements
Weekly	Credit Email	Email to customers
Monthly	Monthly Reports	Business reports
Monthly	Late Interest	Interest on overdue
Monthly	Auto Payroll	Salary slips
Monthly	Auto Commission	Commission for salesmen

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